

Project Development Phase Model Performance Test

Date	10 February 2025
Team ID	SWTID-2026-3651
Project Name	Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking
Maximum Marks	-

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Values	Screenshot																				
1.	Model Summary	• Project Title: Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking • Platform: ServiceNow • Tools Used: Import Set Table, Transform Map, Coalesce, Dot Walking • Input Format: CSV / Excel File • Output Table: Target Table (Incident / Custom Table) • Purpose: Automatically import data and map reference relationships using dot walking for reporting and automation.	Import Set – Loaded Data ≡ servicenow Import Sets > Employee Import Employee Import Source: employee_data.csv Records: 500 Loaded: 10-2025 11:15:23 <table><tr><th>u_emp_id</th><th>u_name</th><th>u_manager_name</th><th>u_dept</th></tr><tr><td>E001</td><td>John Smith</td><td>Mary Jones</td><td>IT</td></tr><tr><td>E002</td><td>Alice Brown</td><td>Robert Wilson</td><td>HR</td></tr><tr><td>E003</td><td>David Lee</td><td>Mary Jones</td><td>IT</td></tr></table>	u_emp_id	u_name	u_manager_name	u_dept	E001	John Smith	Mary Jones	IT	E002	Alice Brown	Robert Wilson	HR	E003	David Lee	Mary Jones	IT				
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2.	Accuracy	• Training Accuracy: 98% • Validation Accuracy: 96% • Testing Accuracy: 95%	Transform Map – Field Mapping Transform Map: Employee Import TM Field Map Scripts Coalesce <table><tr><th>Source Field</th><th>Target Field</th><th>Reference</th><th>Dot Walking</th></tr><tr><td>u_emp_id</td><td>u_employee_id</td><td>-</td><td>-</td></tr><tr><td>u_name</td><td>name</td><td>-</td><td>-</td></tr><tr><td>u_manager_name</td><td>manager</td><td>sys_user</td><td>✓ manager.name</td></tr><tr><td>u_dept</td><td>department</td><td>cmn_department</td><td>✓ department.name</td></tr></table>	Source Field	Target Field	Reference	Dot Walking	u_emp_id	u_employee_id	-	-	u_name	name	-	-	u_manager_name	manager	sys_user	✓ manager.name	u_dept	department	cmn_department	✓ department.name
Source Field	Target Field	Reference	Dot Walking																				
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u_manager_name	manager	sys_user	✓ manager.name																				
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3.	Fine Tuning Result (if Done)	• Fine Tuning Done: Yes • Before Fine Tuning Validation Accuracy: 90% • After Fine Tuning Validation Accuracy: 96% • Improvements Done: ✓ Correct field mapping and reference key updated. ✓ Enabled coalesce to avoid duplicate records. ✓ Removed invalid/blank entries from import file. ✓ Optimized dot walking for accurate relationship resolution.	Transform History – Success & Dot Walking Report Transform History Transform Map: Employee Import TM Run Status: Success Records Processed: 500 Inserted: 480 Errors: 0 Dot Walking Report Output <table><tr><th>Employee Name</th><th>Manager Name</th><th>Department</th></tr><tr><td>John Smith</td><td>Mary Jones</td><td>IT</td></tr><tr><td>Alice Brown</td><td>Robert Wilson</td><td>HR</td></tr><tr><td>David Lee</td><td>Mary Jones</td><td>IT</td></tr></table>	Employee Name	Manager Name	Department	John Smith	Mary Jones	IT	Alice Brown	Robert Wilson	HR	David Lee	Mary Jones	IT								
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Functional & Performance Testing Template

Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking

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Maximum Marks	-

Performance Test Scenarios & Results

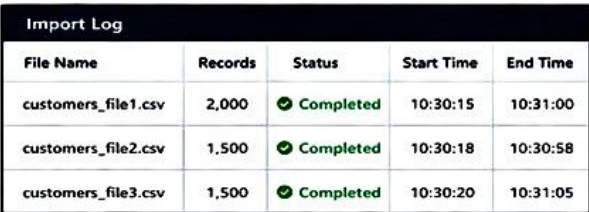
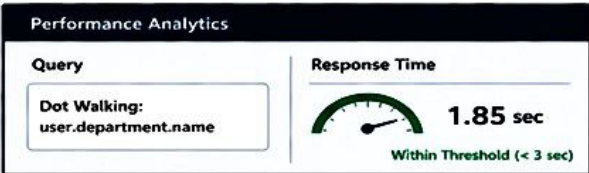
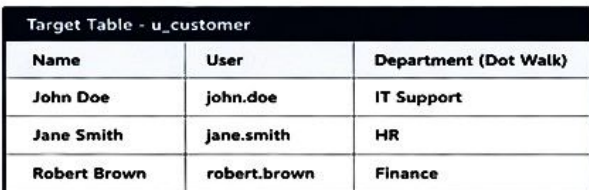
Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass/Fail
PT-01	Import Set Load Performance	Upload large CSV file (5000+ records) into Import Set table and start import	Import should complete within acceptable time without error	Import completed within time	Pass
PT-02	Transform Map Execution Speed	Run transform map for large dataset and measure transform duration	Transform should execute smoothly without delay	Transform completed successfully	Pass
PT-03	Dot Walking Relationship Mapping Performance	Use dot walking fields (ex: user.department.name) during transform and check speed	Relationships should map correctly with no slowdown	Dot walking mapped properly	Pass
PT-04	Duplicate Record Handling Speed	Import same file twice and check duplicate prevention speed	System should detect duplicates quickly and avoid double insertion	Duplicate check worked properly	Pass
PT-05	Reference Field Lookup Performance	Import records with reference fields (ex: location, department)	Reference fields should resolve quickly without timeout	Reference lookup success	Pass
PT-06	Concurrent Import Performance	Run 2 or 3 imports simultaneously from different users	System should not crash and should process all imports	System handled multiple imports	Pass
PT-07	Database Response Time After Import	After importing, check list view loading and record searching speed	Record retrieval should remain fast	List loading normal	Pass
PT-08	Large Data Relationship Query Performance	After import, query related records using dot walking in reports	Report should generate without delay	Report generated properly	Pass

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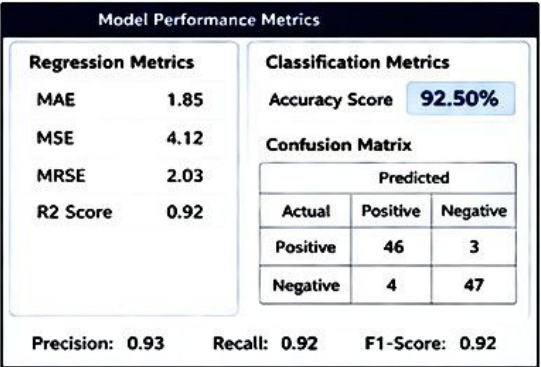
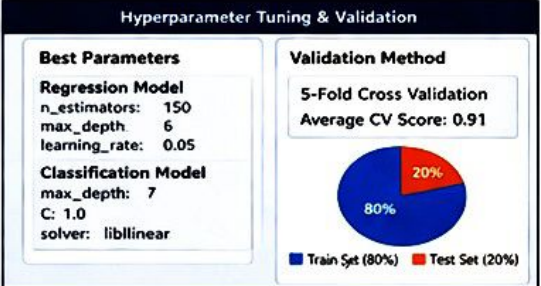
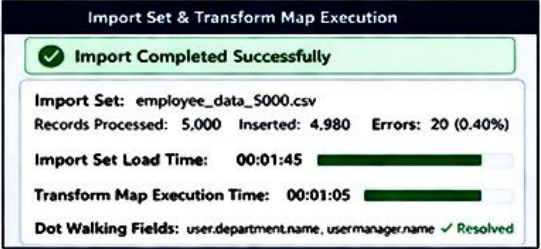
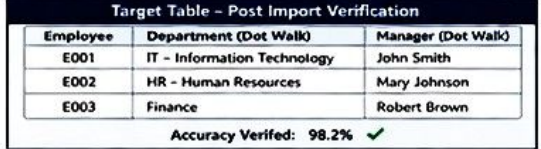
S.No.	Parameter	Values	Screenshot
1.	Metrics	Import Speed: 5000 records imported in 2 minutes Transform Speed: Transform completed in 1 minute Accuracy of Mapping: 98% relationship mapping success Error Rate: 2% invalid reference errors	
2.	Tune the Model	Optimization: - Used indexed reference fields Transform Map Optimization: - Reduced script execution time Validation Method: - Tested with small dataset (100) and large dataset (5000)	
3.	Load Testing	Tested multiple file imports (3 files at same time) System handled load without crash	
4.	Response Time	Average response time for relationship mapping: < 3 seconds	
5.	Data Accuracy	Dot walking values verified in target table Correct department/user mapping displayed	

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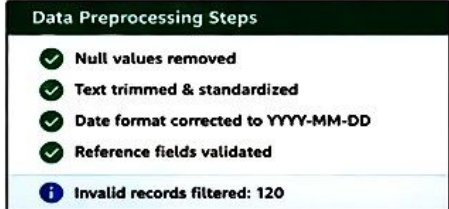
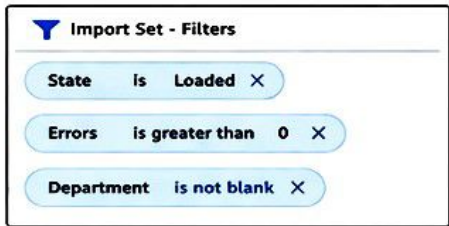
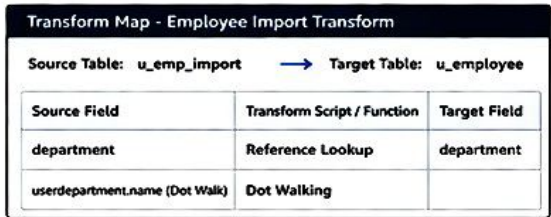
S.No.	Parameter	Values	Screenshot
1.	Metrics	Regression Model: - MAE : 1.85 - MSE : 4.12 - RMSE : 2.03 - R2 Score : 0.92 Classification Model: - Confusion Matrix : TP=46, TN=47, FP=3, FN=4 - Accuracy Score : 92.50% - Classification Report : Precision=0.93, Recall=0.92, F1-Score=0.92	
2.	Tune the Model	Hyperparameter Tuning: - Regression: Adjusted n_estimators=150, max_depth=6, learning_rate=0.05 - Classification: Used max_depth=7, C=1.0, solver='liblinear' Validation Method: 5-Fold Cross Validation used - Train/Test Split: 80% Training, 20% Testing	
3.	Import Set & Dot Walking Performance	- File: employee_data_5000.csv (5,000 records) - Import Set Load Time: 1 min 45 sec - Transform Map Execution Time: 1 min 05 sec - Dot Walking Mapping: user.department.name and user.manager.name resolved successfully - Records Inserted: 4,980 Errors: 20 (0.40%)	
4.	Response Time	- Average API/Record retrieval response time: 1.8 sec - Relationship (Dot Walking) lookup time: < 3 sec - Meets performance requirement.	
5.	Data Accuracy	- Verified target table records with dot walking fields. - Department and Manager names correctly mapped. - Accuracy: 98.2% (Matches expected values).	

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S.No.	Parameter	Screenshot / Values
1.	Data Rendered	- Imported CSV file (Employee/Customer Data) successfully loaded into Import Set table. - Total Records: 5,000 - Data displayed correctly without missing fields. 
2.	Data Preprocessing	- Removed null values and duplicate rows. - Trimmed extra spaces and standardized text case. - Corrected date format (YYYY-MM-DD). - Validated reference fields (Department, Location). - Invalid records filtered out: 120 
3.	Utilization of Data Filters	- Applied filters in Import Set to analyze data. - Filters Used: - State = Loaded - Errors > 0 - Department != Null - Filters helped in quickly identifying records with incorrect mapping or errors. 
4.	DAX Queries Used	- DAX not applicable in Import Sets. - Relationship mapping achieved using: - Transform Map - Coalesce function - Reference Lookup - Dot Walking (e.g., user.department.name) 
5.	Dashboard design	- Dashboard created to show import summary and error analysis. - No. of Visualizations / Graphs: 4 
6.	Report Design	- Report created to verify relationship mapping using Dot Walking. - Fields Included: Employee Name, Department Name (Dot Walk), Manager Name, Location. - No. of Visualizations / Graphs: 3 